Recap Checkpoint 2 — after 1.3 (Exchanging Data) — cumulative 1.1 + 1.2 + 1.3 — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Total = 36 marks. Award marks for valid alternatives in line with OCR positive-marking.

Q1 — [2] (AO1) — *1.1.1* - MAR holds the **address** of the memory location to be read from / written to (1). - MDR holds the **data or instruction** fetched from, or to be written to, memory (1).

Q2 — [3] (AO1) — *1.1.1* - Cache is **fast memory close to / on the CPU** holding **frequently/recently used instructions and data** (1)… - …so the CPU avoids slower fetches from main memory, reducing access time (1). - Difference (1): **L1 is smaller but faster (and closest to the core); L3 is larger but slower** (and often shared between cores).

Q3 — [3] (AO1) — *1.2.1* - The key press generates an **interrupt** / sets an interrupt flag (1). - At the end of the current FDE cycle the CPU **saves the current state to the stack** and **runs the ISR** for the keyboard (1). - After the ISR completes, the **saved state is restored** and the interrupted program resumes (1).

Q4 — [3] — *1.2.2* (a) Bytecode is **platform-independent / portable** (1) — the same file runs on any device with the appropriate virtual machine, so it need not be recompiled per platform (1). *(AO2)* (b) Any one (1): it **runs slower** than native machine code (translation/JIT overhead at run time); the **VM must be installed**. *(AO2)*

Q5 — [3] (AO1) — *1.3.1* - Hashing applies a **one-way function** to input to produce a **fixed-length hash/digest** that cannot feasibly be reversed (1). - Reason 1 (1): if the database is stolen, the **original passwords are not exposed** (irreversible). - Reason 2 (1): the stored hash can still be **verified** at login by hashing the entry and comparing, without storing the plaintext.

Q6 — [3] — *1.3.1* (a) **5R 7G 4R** (accept equivalent notation such as `R5 G7 R4`, or value/run pairs) (2 marks; 1 mark if one run count wrong). *(AO2)* (b) An image with **no long runs of identical pixels** — e.g. a **photograph / highly detailed or noisy image** (1) — RLE may even **increase** size because every short run still needs a count + value. *(AO2) Check: 5 + 7 + 4 = 16 pixels ✓.*

Q7 — [4] — *1.3.1* (a) **Symmetric** uses **the same (single) key** to encrypt and decrypt; **asymmetric** uses a **key pair — public to encrypt, private to decrypt** (or vice versa for signing) (2; 1 per type). *(AO1)* (b) With asymmetric encryption the recipient's **public key can be shared openly** (1), so the sender encrypts with it and only the holder of the matching **private key can decrypt** — **no secret key ever needs to be exchanged**, solving key distribution for first contact (1). *(AO2)*

Q8 — [4] — *1.3.2* (a) Every field/attribute is **atomic** (single value) with **no repeating groups** (and a primary key) (1). *(AO1)* (b) It is not in 3NF because of **transitive / non-key dependencies**: non-key fields depend on other non-key fields (`CustomerName`/`CustomerEmail` on `CustomerID`; `ProductName` on `ProductID`) (1). Resulting tables (2): *(AO2)* - `Customer(CustomerID, CustomerName, CustomerEmail)` - `Product(ProductID, ProductName)` - `Order(OrderID, CustomerID*, ProductID*)` *(Award 1 for identifying the transitive dependency, 1 for the Customer/Product split, 1 for the linking Order table with foreign keys.)*

Q9 — [4] (AO2) — *1.3.2*

```
SELECT Surname, Cost
FROM Booking
WHERE Town = 'York' AND Nights > 3
ORDER BY Cost DESC;
```

- `SELECT Surname, Cost` (1)
- `FROM Booking` (1)
- `WHERE Town = 'York' AND Nights > 3` (1)
- `ORDER BY Cost DESC` (1)

Q10 — [4] (AO1) — *1.3.3* (a) **Application** → **Transport** → **Network (Internet)** → **Link (Data Link/Network Access)** (2; 1 mark if one pair out of order). (b) The **IP address is added at the Network layer** (1); the **Transport layer splits data into packets and reassembles them** (TCP) (1).

Q11 — [3] (AO3) — *synoptic 1.3.4* Up to 3 from a reasoned evaluation: - Moving work **client-side (e.g. JavaScript validation/rendering)** reduces **server load and round-trips**, so pages feel faster / scale to more users (1). - Trade-off: client-side code **runs on the user's device** and can be **disabled/bypassed**, so security-critical checks must stay server-side (1). - Better **indexing / using meta-data, relevant content and inbound links** improves how a search engine's algorithm (e.g. **PageRank**) ranks the site, raising visibility (1). *Indicative answer links faster pages + stronger ranking signals to more traffic, while noting client-side limits.*